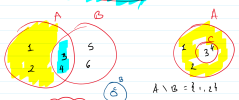
Est. SEP 16
miércoles, septiembre 16, 2020 2:13 PM

$\{x : x \in A \wedge x \notin B\}$

Si $\exists\, C \subseteq A \Rightarrow A \setminus B = A \setminus C = \{x : x \in A \wedge x \notin C\}$



$A \setminus B = \{1, 2\}$ $A \setminus C = \{1, 2\}$

① Contubern $C = A \cap B$ probaré que $A \setminus B = A \setminus C$

Sea $x \in A \setminus B \Rightarrow x \in A \wedge x \notin B$

$\Rightarrow x \in A \wedge x \notin C$

$x \in A \wedge x \notin C \Rightarrow x \in A \setminus C$

$\Rightarrow A \setminus B \subseteq A \setminus C$



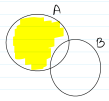
② Sea $x \in A \setminus C \Rightarrow x \in A \wedge x \notin C$

A priori yo no se si $x \in B$, sup. $x \in B \Rightarrow x \in A \cap B = C$ ↯

$\Rightarrow x \in A \wedge x \notin B \Rightarrow x \in A \setminus B \Rightarrow A \setminus C \subseteq A \setminus B$.

Def: la diferencia $A \setminus B$ es el conjunto

$\{x : x \in A \wedge x \in B\}$



$\mathcal{U} = \mathbb{R}$

- $\mathbb{Z} \setminus \mathbb{N} = \{-1, -2, -3, \dots\}$
- $\mathbb{R} \setminus \mathbb{Z} =$  $= (0,1) \cup (-1,0) \cup (1,2) \cup (-2,-1) \cup \dots$

① $\mathbb{Z}^c$ $\mathbb{Z} \subseteq \mathbb{Q}$ $= \bigcup_{n \in \mathbb{Z}} (n, n+1)$

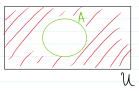
$\mathbb{Q}^c$ $\mathbb{Q}^c \subseteq \mathbb{Z}^c$

- $\mathbb{R} \setminus \mathbb{Q} =$ ..... $= \mathbb{I}$ irracionales.

② $\mathbb{R} \setminus \mathbb{Q} \subseteq \mathbb{R} \setminus \mathbb{Z}$

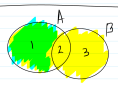
Def: $\mathcal{U}$ un universo de contexto $A \subseteq \mathcal{U}$.

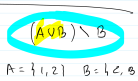
$A^c = \mathcal{U} \setminus A = \{x : x \notin A\}$



A^c $A \cup A^c = \mathcal{U}$

$A \cap A^c = \emptyset$.





$A = \{1, 2\}$ $B = \{2, 3\}$

$A \cup B = \{1, 2, 3\}$

$(A \cup B) \setminus B = \{1\} \neq A$.

$(A \cup B) \setminus B = A \setminus B$

$\subseteq$ Sea $x \in (A \cup B) \setminus B \Rightarrow x \in A \cup B \wedge x \notin B$

$\Rightarrow (x \in A \vee x \in B) \wedge x \notin B$ $(p \vee F \equiv p)$

$\Rightarrow (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin B)$ (F)

$\Rightarrow (x \in A \wedge x \notin B) \Rightarrow x \in A \setminus B$.

$\supseteq$ $x \in A \setminus B \Rightarrow x \in A \wedge x \notin B \Rightarrow$

$(x \in A \vee x \in B) \wedge x \notin B$

$\Rightarrow x \in A \cup B \wedge x \notin B$

$\Rightarrow x \in (A \cup B) \setminus B$.

Ej $A \not\subseteq B \iff A \setminus B \neq \emptyset$

$\Rightarrow$ Hipótesis: $A \not\subseteq B$ Mostraré: $A \setminus B \neq \emptyset$.

Hip: $\exists x : x \in A \wedge x \notin B$ Mostraré: $\exists\, x \in A \setminus B$

$\exists x : x \in A \wedge x \notin B$

$\neg(A \not\subseteq B) \equiv A \subseteq B$ ↯

$\forall x : x \in A \Rightarrow x \in B$

$\neg$ $\forall x : x \notin A \vee x \in B$

$\exists x : x \in A \wedge x \notin B$

$\Leftarrow$ Hipótesis $A \setminus B \neq \emptyset$ Mostraré $A \not\subseteq B$

$\exists x : x \in A \setminus B \Rightarrow \exists x : x \in A \wedge x \notin B$

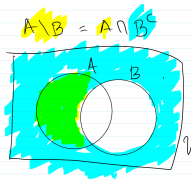
$\Rightarrow \neg(\forall x : x \notin A \vee x \in B)$

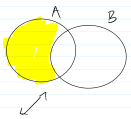
$\Rightarrow \neg(\forall x : x \in A \Rightarrow x \in B)$

$\equiv \neg(A \subseteq B) \Rightarrow A \not\subseteq B$

F o V: $A \setminus B = A \cap B^c$

Pruebo





$\subseteq$ Sea $x \in A \setminus B \Leftrightarrow x \in A \wedge x \notin B \Leftrightarrow x \in A \wedge x \in B^c \Leftrightarrow x \in A \cap B^c$

Teo: $(A \cap B) \cup (A \cap B^c) = A$



Prof:

$(A \cap B) \cup (A \cap B^c)$

$[A \cap (A \cap B^c)] \cup [B \cap (A \cap B^c)]$

$= [(A \cap A) \cap B^c] \cup [$